

	discharged.
8(a)	The half life of 87.7 years is long enough to sustain power generation for the lifespan of 14 years of Perseverance, without rapid decay. This makes it suitable.
8(b)	α -particles provide strong ionisation and lose energy over a short distance. This results in efficient heat generation by kinetic energy absorption within a small volume, most suitable for an RTG.
8(c)	α -particles emitted by plutonium produce strong ionisation. If it is absorbed by living organisms, their living cells would be exposed to this radiation. This might cause mutations or other harmful effect for the cells, which might lead to health hazards or even death for the organisms.
8(d)(i)	${}_{94}^{238}\text{Pu} \rightarrow {}_{92}^{234}\text{U} + {}_2^4\alpha$
8(d)(ii)	Binding energy of Pu, $E_1 = (238)(7.568 \text{ MeV}) = 1801.184 \text{ MeV}$ Binding energy of U, $E_2 = (234)(7.601 \text{ MeV}) = 1778.634 \text{ MeV}$ Binding energy of α, $E_3 = (4)(7.074 \text{ MeV}) = 28.296 \text{ MeV}$ Energy emitted = $E_2 + E_3 - E_1 = 5.746 \text{ MeV}$ $= (5.746)(1.60 \times 10^{-13}) \text{ J}$ $= 9.19 \times 10^{-13} \text{ J}$

8(d)(iii)	Mass of plutonium dioxide = 4.9 kg = 4900 g Molar mass of plutonium dioxide = 276 g mol ⁻¹ Number of nuclei, $N = \left(\frac{4900}{276}\right)(6.02 \times 10^{23})$ $= 1.07 \times 10^{25}$
8(d)(iv)	Decay constant $\lambda = \frac{\ln 2}{t_{1/2}} = \frac{\ln 2}{(87.7)(365)(24)(3600)} = 2.506 \times 10^{-10} \text{ Bq}$ Rate of decay $A = \lambda N = (2.506 \times 10^{-10})(1.07 \times 10^{25}) = 2.68 \times 10^{15} \text{ s}^{-1}$ From (d)(ii), energy emitted in a decay = $9.19 \times 10^{-13} \text{ J}$ So power output = $(2.68 \times 10^{15})(9.19 \times 10^{-13}) = 2460 \text{ W} = 2500 \text{ W}$
8(d)(v)	$238 - 206 = 32$ Each α -decay reduces the nucleon number by 4. $32 \div 4 = 8$ So number of α -decay = 8 Each α -decay reduces the proton number by 2. $8 \times 2 = 16$ $94 - 82 = 12$ $16 - 12 = 4$ Each β -decay increases the proton number by 1. So number of β -decay = 4
8(e)(i)	An exponential decrease is one where each decay decreases the number of nuclei by a fixed fraction of the previous number.
8(e)(ii)	Each year its power decreases by 0.79%, i.e. fraction of power remained = $\frac{100 - 0.79}{100} = 0.9921$ After 14 years, output power = $(110 \text{ W})(0.9921)^{14}$ $= 98.4 \text{ W} = 98 \text{ W}$
8(f)	Difference in temperature $\Delta T = 1273 - 573 = 700 \text{ K}$ $\Delta S = 240 - (-200) = 440 \mu\text{V K}^{-1}$ e.m.f. = $\Delta S \Delta T = (440)(700) = 308000 \mu\text{V} = 0.308 \text{ V} = 0.31 \text{ V}$
8(g)	For Mars, $g_M = \frac{GM_M}{r_M^2}$ --- (1) For Earth, $g = \frac{GM}{r^2}$ --- (2) $\frac{(1)}{(2)} :$ $\frac{g_M}{g} = \left(\frac{M_M}{M}\right)\left(\frac{r}{r_M}\right)^2 = \left(\frac{1}{9.30}\right)(1.88)^2 = 0.380$ $g_M = (0.380)(9.81) = 3.73 \text{ N kg}^{-1} = 3.7 \text{ N kg}^{-1}$
8(h)(i)	RTG provides 110 W continuously, to be stored in onboard batteries. When MOXIE is operated for a few hours, most likely when the other onboard operations are idle, the required power of 300 W could be drawn from the charged batteries.

8(h)(ii)	From Fig. 8.2, time taken $t = 12400$ s Energy required to run MOXIE, $E = Pt = (300)(12400) = 3.72 \times 10^6$ J Time taken for RTG to generate this energy, t_1 is given by $t_1 = \frac{E}{110} = \frac{3.72 \times 10^6}{110} = 33818 \text{ s}$ $= 9.39 \text{ hours}$ $= 9.4 \text{ hours}$	
8(h)(iii)	The amount in 5.37 g of oxygen is given by $n = \frac{5.37}{32} = 0.1678 \text{ mol}$ Temperature $T = -60^\circ\text{C} = (-60 + 273.15) \text{ K} = 213.15 \text{ K}$ Using $pV = nRT$, $V = \frac{nRT}{p} = \frac{(0.1678)(8.31)(213.15)}{610}$ $= 0.487 \text{ m}^3 = 0.49 \text{ m}^3$	